

Week 2 – Practice Session

September 14–18, 2026

In every problem, $x, y \in \mathbb{R}$.

1 Linear Systems of Equations

1. Solve the following systems of equations using the substitution method!

a) $\begin{cases} 2x + y = 4 \\ 4x + 3y = 6 \end{cases}$

b) $\begin{cases} x + 12y = 35 \\ 3x - 8y = 17 \end{cases}$

c) $\begin{cases} 7x - 2y = 12 \\ 4x + 6y = 14 \end{cases}$

d) $\begin{cases} 2x + 7y - 78 = 0 \\ 2x + 10y - 95 = 16 \end{cases}$

2. Solve the following systems of equations using the comparison method!

a) $\begin{cases} x + 2y = 17 \\ x - y = 11 \end{cases}$

b) $\begin{cases} 5x - 2y = 19 \\ 3x - 10y = -15 \end{cases}$

c) $\begin{cases} |x| + y = 3 \\ |x - 1| = y + 2 \end{cases}$

3. Solve the following systems of equations using the method of introducing new variables!

a) $\begin{cases} \frac{10}{x-5} + \frac{1}{y+2} = 1 \\ \frac{25}{x-5} + \frac{3}{y+2} = 2 \end{cases}$

b) $\begin{cases} 2(x+y) + 4(x-y) = 3 \\ 3(x+y) + 6(x-y) = 4.5 \end{cases}$

2 Word Problems

1. Benches are being placed in a physics classroom. If two students sit on each bench, there will not be enough seats for 8 students. However, if three students sit on each bench, 7 seats will remain empty. How many benches are there in the classroom? How many students are there in the class?
2. We are collecting money for a trip. If each participant contributes 75 HUF, we are still 440 HUF short of the required amount. However, if everyone contributes 80 HUF, there will be 440 HUF left over. How many people are participating in the trip?
3. In sixteen years, the father will be twice as old as his son. How old are they now if the father was six times as old as his son four years ago?
4. A 75% mixture is prepared from alcohol solutions with concentrations of 84% and 70%. Then, 5 cm^3 of the first type and 135 cm^3 of the second type are added to the mixture. The resulting mixture has a concentration of 72%. How many cm^3 of each type were used initially?
5. A boy has the same number of sisters as brothers, and his sister has half as many sisters as brothers. How many brothers and sisters are there in the family?
6. The graph of a linear function passes through the points $(-7; 9)$ and $(0; -1)$. Give the rule of the linear function in the form $y = mx + b$!

3 Quadratic Systems of Equations

a) $\begin{cases} x + y = 9 \\ xy = 8 \end{cases}$

b) $\begin{cases} 3x + 4y = -18 \\ xy = 6 \end{cases}$

c) $\begin{cases} 2y + 4x = x^2 + 2 \\ y - x = 1 \end{cases}$

d) $\begin{cases} \frac{y-4}{x+1} = 3 \\ (y-4) \cdot (x+1) = 12 \end{cases}$

e) $\begin{cases} x^2 + y^2 = 10x - 2y + 8 \\ x^2 + y = 10x - 22 \end{cases}$

f) $\begin{cases} x + y^2 = 7 \\ xy^2 = 12 \end{cases}$

g) $\begin{cases} x^2 + y^2 + x + y = 18 \\ x^2 - y^2 + x - y = 6 \end{cases}$

h) $\begin{cases} 2\sqrt{x} + \sqrt{y} = 5 \\ x + \sqrt{y} = 2 \end{cases}$

i) $\begin{cases} (x+1)^2 + (y+3)^2 = 25 \\ (x-2)^2 + (y+9)^2 = 10 \end{cases}$